DE–439

Sub. Code
11

DISTANCE EDUCATION

M.C.A. (S) DEGREE EXAMINATION, MAY 2017.

First Semester

PRINCIPLES OF INFORMATION TECHNOLOGY

(2005 to 2010 CY)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) What are the basic operations of computing? Explain. (10)

   (b) Explain different kinds of computers. (10)

2. (a) Explain development in communication technology. (10)

   (b) Describe interactive devices. (10)

3. (a) Explain formatting a word document. (10)

   (b) Write the features of spreadsheet. (10)

4. (a) Write the capabilities of communication software. (10)

   (b) Discuss on copyright. (10)

5. (a) Write the offerings of online services. (10)

   (b) Describe important aspects of cyber ethics. (10)

6. (a) Explain lossless and lossy compressions. (10)

   (b) How to take care of diskettes? Explain. (10)

7. (a) Describe the different types of database organisation. (10)

   (b) Write the features of DBMS. (10)

8. Describe the phases of system analysis and design. (20)

DE–440

Sub. Code
12

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2017.

First Semester

DIGITAL COMPUTER ORGANIZATION

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Convert the following hexadecimal numbers to equivalent decimal numbers :

   (i) 1A 5B

   (ii) 3A9

   (iii) 6B

   (iv) 6AC

   (b) Convert the following decimal number to Binary, Octal and hexadecimal 4398.

2. (a) Subtract the $(22)_{10}$ from $(55)_{10}$ using 2's complement method.

   (b) Perform the following subtractions and check the result by converting the binary numbers to decimal.

   (i) 1101–1001

   (ii) 11001–10110

3. (a) What are the basic laws of Boolean algebra? Explain.

   (b) Explain the working principle of an Encoder and draw the circuit diagram.

4. (a) What are sum of products form and product of sum form of logic expressions? Explain with an example.

   (b) Explain Quine Mck Luskey tabulation method with an example.

5. (a) Explain the half-adder and full-adder functions with truth table and circuit diagram.

   (b) With block diagram, explain the 4-bit parallel binary adder.

6. (a) With a neat circuit diagram and truth tables explain the working of master slave flip-flop.

   (b) What are static and dynamic RAM? Explain.

7. (a) What are the functions of a shift register? What are its various types? Explain.

   (b) Discuss the working principle of a BCD counter.

8. (a) What do you understand by 'peripheral'? Explain with examples.

   (b) Use a block diagram to explain DMA controller. How DMA transfer is implemented?

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Sub. Code
13

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2017.

First Semester

C AND DATA STRUCTURES

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Explain various data types in C. (10)

   (b) Describe printf() and scanf() function with example program. (10)

2. (a) Write short notes on different types of array with example program. (10)

   (b) What is recursion? Explain with suitable example. (10)

3. (a) Briefly explain structure in C. (10)

   (b) Write a program to copy the contents of one file into another. (10)

4. (a) Briefly explain stack operations. (10)

(b) Convert the given Infix expression to postfix expression using Stack and show the details of Stack at each step of conversion.

Expression: $(a + b * c\hat{}d) * (e + f / g)$.

Note : ^ indicates exponent operator. (10)

5. (a) Describe the structure of doubly linked list. (10)

(b) Give stepwise procedure for in-order and post-order traversal of Binary Tree. (10)

6. (a) Explain different types of decision control statements in C with suitable example. (10)

(b) Describe the library function and macros. (10)

7. (a) Describe Binary Search with the help of an example. (10)

(b) Explain any two applications of queue. (10)

8. (a) Explain any five string functions with suitable example. (10)

(b) Explain the different kinds of loops available in C with examples. (10)

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Sub. Code
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DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2017.

First Semester

OFFICE AUTOMATION

(2005 to 2010 Calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Give a detailed note on managing files with windows explorer and working with multiple Windows. (10)

   (b) Explain in detail about how to create, open and save the file in Microsoft Word with relevant dialog boxes for each of them. (10)

2. (a) Discuss in detail about the various keys used for editing in Microsoft Word. (10)

   (b) How can the user handle tables in Microsoft Word? (10)

3. (a) What are the various options available for printing a document in Microsoft Word? Explain them. (10)

   (b) Write down the step by step procedure to implement mail merge feature in Microsoft Word with corresponding dialog boxes. (10)

4. (a) What is a formula bar in Microsoft Excel? Explain in detail about the use of formula bar in Microsoft Excel. (10)

   (b) How can the user insert and resize rows and columns in Microsoft Excel? (10)

5. (a) Write a note on conditional formatting in Microsoft Excel. (10)

   (b) Explain how the user can create various kinds of charts in Microsoft Excel with relevant dialog boxes. (10)

6. (a) What are the various views available in Microsoft Excel? Explain. (10)

   (b) Write a detailed note on auto layouts and handouts in Microsoft Excel. (10)

7. (a) Discuss how the user can create tables using primary key in Microsoft Access. (10)

   (b) Write a detailed note on the various operations that can be performed on fields in Microsoft Access. (10)

8. (a) Discuss about how the user can create and print reports in Microsoft Excel. (10)

   (b) Explain how the user can combine the data and charts in excel with word documents. (10)

DE–9532

Sub. Code
15

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2017.

First Semester

LAB I – DATA STRUCTURES USING "C"

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Examiner should select and give ONE question to each candidate by LOT system.

1. (a) Write a C program to find the roots of a quadratic equation.

   (b) Write a C program for linked list implementation of queue operations.

---------- Cut here ----------

2. (a) A man is paid at the hourly rate of Rs. 25 per hour for the first 50 hours worked. Thereafter, over time is paid at 2 time the hourly rate for the next 25 hours and 3 time the hourly rate for further hours worked. Write a C program to input the number of hours worked per week, calculate and print his gross weekly wage.

   (b) Write a C program to implement insert and delete operations on linked list structure.

3. (a) Write a C program to find the product of two matrices.

   (b) Write a C program to implement PUSH and POP operations on stack.

---------- Cut here ----------

4. (a) Write a C program to implement all string operations using switch statement.

   (b) Write a menu driven program in C to sort the given set of numbers in ascending and descending order.

---------- Cut here ----------

5. (a) Write a C program to find a word is palindrome or not.

   (b) Write a C program to evaluate the given mathematical expression using stack.

---------- Cut here ----------

6. (a) Write a C program to read 10 values to an array variable. Use pointers to locate and display each value.

   (b) Write a C program for linked list implementation of queue operations.

---------- Cut here ----------

7. (a) Write a C program to copy the contents of one file to another file.

   (b) Write a C program to find an element using binary search.

8. (a) Write a C program to find the factorial of a given number using recursion.

(b) Write a C program to implement insert and delete operations on queue using array concept.

DE–9533

Sub. Code
16

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2017.

First Semester

LAB II — MS OFFICE AND INTERNET

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Examine should select and give ONE question to each candidate by LOT system.

1. (a) Prepare a news report using two columns, insert a picture in the first column and make the text flow around it.

   (b) Create a HTML page that receives employee name, age, street, city, pin code. Receive male/female and married/unmarried details through option buttons. Receive hobbies through check boxes.

---------- Cut here ----------

2. (a) Create a worksheet in Excel for electricity bill :

Consumer Number	User Name	Previous type	Current Reading	units Reading	Bill

   (i) Add data for 10 customers with type D for domestic user and type I for industrial user.

   (ii) Find units column for each customer and calculate bill using different slab rates (Assume your own slab rates)

   (iii) All the consumers should be charged a minimum bill of Rs.50 even if their bill amount is below Rs. 50.

   (b) Create HTML page to display the pictures of elephant and monkey. Link to another page when user clicks over the picture. The linked page should display features of the animal clicked.

3. (a) Create employee table with the fields; empno, name, age, sex, street, city, pin, salary.

   (i) Add data for 10 employees

   (ii) Write a query to display all the male employees whose salary is between 10000 and 50000 and living in city "Coimbatore".

   (iii) Show all the records in the table for the city "Trichy" by filtering.

   (b) Create a HTML page that plays video and audio files.

---------- Cut here ----------

4. (a) Prepare the first page of MCA practical record not book with picture insertion and alignment.

   (b) Develop a complete set of web pages to describe your skills in various areas.

---------- Cut here ----------

5. (a) Create a worksheet with the following columns :

   Salesman number, salesman name, city,

   Product name, sale amount.

   Add three records for five different salesman who have carried out sales of different products in different cities. Find the following :

   (i) City wise Total Sales

   (ii) Salesman wise Total Sales

   (iii) Product wise Total Sales

   (iv) Grand Total sales.

   (b) Create HTML page that divides the screen space into three frames (one vertical and two horizontal of equal size) Display three different files in them with scrolling titles relevant to their contents.

6. (a) Create two tables SALES 1, SALES 2 and join them to produce a third table SALES 3.

   (b) Create HTML page to display ten courses and their highlighted features, conducted by an educational institution in a colorful format.

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Sub. Code
21

DISTANCE EDUCATION

M.C.A. (S) DEGREE EXAMINATION, MAY 2017.

Second Semester

OBJECT ORIENTED PROGRAMMING AND C++

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

($5 \times 20 = 100$)

All questions carry equal marks.

1. (a) List out some benefits of object oriented programming and describe the applications of OOPS. (10)

   (b) Describe the different types of tokens used in C++. (10)

2. (a) Discuss the different types of 'If' constructs. (10)

   (b) Describe how to declare multidimensional array. (10)

3. (a) Differentiate between while...loop and do...while statement. (10)

   (b) Explain the general format for structure used in C++. (10)

4. (a) Explain why we need of function concept. (10)

   (b) How do you apply pointer to user defined function? (10)

5. (a) Explain the concept of overloaded functions. (10)

   (b) What is meant by virtual function? Explain it with an example. (10)

6. (a) Explain the concept of constructor with an example. (10)

   (b) Discuss operator overloading concept with an example. (10)

7. (a) What is inheritance? Describe in detail about hybrid inheritance with an example. (10)

   (b) Difference between virtual function and polymorphism. (10)

8. (a) Discuss the classes for file stream operations. (10)

   (b) How do you perform command line argument? Give an example. (10)

DE–444

Sub. Code
22

DISTANCE EDUCATION

M.C.A. (S) DEGREE EXAMINATION, MAY 2017.

Second Semester

SYSTEMS ANALYSIS AND DESIGN

(2005 to 2010 Calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

($5 \times 20 = 100$)

1. (a) Discuss about system development life cycle. (12)

   (b) Explain the skills of an analyst. (8)

2. (a) What do you mean by feasibility study? Explain. (10)

   (b) How does the analyst acquire academic and personal skills? Explain. (10)

3. (a) Write a note on oral presentation. (10)

   (b) Explain the classifications of costs and benefits. (10)

4. (a) Write a note on decision trees. (10)

   (b) What is data flow diagram? Explain. (10)

5. (a) How is logical design differs from physical design? Explain. (10)

   (b) Explain the requirements of forms design. (10)

6. (a) Explain the major development activities in structured design. (10)

   (b) Write a note on HIPO and IPO charts. (10)

7. (a) Discuss about Quality Assurance. (10)

   (b) Explain the types of data structure. (10)

8. (a) Write a note on training methods. (10)

   (b) Explain the procedure for hardware and software selection. (10)

DE–445

Sub. Code
23

DISTANCE EDUCATION

M.C.A. (S) DEGREE EXAMINATION, MAY 2017.

Second Semester

VISUAL PROGRAMMING — I

(2005 to 2010 CY)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) What are custom components? Explain in detail. (10)

   (b) What are event procedures? Explain with an example. (10)

2. (a) What are the properties, methods and events of Text Box Control? (10)

   (b) Describe format functions in VB with examples. (10)

3. (a) What are procedures? Explain its types. (10)

   (b) Discuss about properties and methods of Flex Grid control. (10)

4. (a) How can you create Menus? Explain with an example. (10)

   (b) How can you draw geometric objects in VB? Explain. (10)

5. (a) Write a VB program to display the drives, directories and files in your system. (10)

   (b) Explain OLE automation. (10)

6. (a) Explain If...Then..Else statement with example coding. (10)

   (b) What is control array? How can you add and remove a control in a control array? (10)

7. (a) Explain the following :

   (i) Do Events. (5)

   (ii) Error Trapping. (5)

   (b) Describe common properties of forms. (10)

8. (a) Explain the following :

   (i) Clipboard. (5)

   (ii) Shell function. (5)

   (b) Describe the events and methods of Data Control. (10)

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Sub. Code
24

DISTANCE EDUCATION

M.C.A.(S) DEGREE EXAMINATION, MAY 2017.

Second Semester

COMPUTER GRAPHICS

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) How mouse acts as a good input device for graphics? (10)

   (b) Describe an algorithm for circle generation. (10)

2. (a) Explain the simple DDA algorithm for line generation. (10)

   (b) Discuss the applications of computer graphics. (10)

3. (a) Give the matrix representation for scaling and rotation. (10)

   (b) Describe the Sutherland Hodgman algorithm for clipping. (10)

4. (a) Explain the concept of composite transformations. (10)

   (b) Discuss an algorithm for point clipping. (10)

5. (a) Mention the concepts of three dimensional transformations. (10)

   (b) How to obtain a mirror reflection for a three dimensional object? (10)

6. (a) Discuss the viewing process of a three dimensional object. (10)

   (b) Explain the Z-buffer algorithm. (10)

7. (a) Define projection. Discuss on its types. (10)

   (b) How depth comparison is done with respect to a three dimensional object? (10)

8. (a) Discuss the components of user interface. (10)

   (b) Give the importance of feed back in user interface design. (10)

DE–9534

Sub. Code
25

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2017.

Second Semester

LAB III — GRAPHICS USING C++

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Examiner has to choose ONE question for each candidate by LOT system.

Break-up of marks

Record Note Book : 5
Algorithm/Flow chart : 15
Program : 35
Debugging and Execution : 35
Result : 10
Total : 100

1. Create a base class shape, which contains the common elements of circle, triangle and rectangle, from which you derive three classes called ball, rect and tria. Include two member functions set() to set coordinates and draw() which draws them. Write a program to exercise this class using graphic-mode graphic functions.

2. Write a C++ Program to draw a straight line using simple DDA technique.

---------- Cut here ----------

3. Write a C++ Program to draw non linear object ellipse.

---------- Cut here ----------

4. Write a C++ Program to draw non linear objects arcs and curves.

---------- Cut here ----------

5. Write a C++ Program to show graphical transformation translation of a two dimensional objects.

---------- Cut here ----------

6. Write a C++ Program to show graphical transformation Rotation on two dimensional object.

---------- Cut here ----------

7. Write a C++ Program to show graphical transformation Scaling on three dimensional object.

8. Write a C++ Program to draw circle using Bresenham's algorithm.

---------- Cut here ----------

9. Write a C++ Program to manipulate the graphical objects using SHADING technique.

---------- Cut here ----------

10. Draw any 3-D object and perform animation on it.

DE–9535

Sub. Code
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DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2017.

Second Semester

LAB — IV (VISUAL BASIC)

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Examiner has to select ONE question and give it to the candidate by LOT system.

1. (a) Write a VB project that receives a foreign currency value selected from a list box and converts it into equivalent rupees.

   (b) Write a VB project using ActiveX control to create a text box that accepts only numeric value. Provide the following properties for the text box :

   Back color, Fore color, Text.

---------- Cut here ----------

2. (a) Write a VB project to create a screen saver that displays a list of pictures with one second pause in between successive pictures.

   (b) Write a VB project using AX DLL or EXE, add a class module that would perform the following functions :

   (i) Test whether the given number is Armstrong or not.

   (ii) Find the factorial of the given number.

3. (a) Write a VB project that receives a year number from a text box and month name from list box and displays number of days in the given month. Take care of leap years. Use lost focus event for list box.

(b) Write a VB project for a blood bank that maintains a list of donors with address and their blood group. Provide the following reporting features :

(i) Display all donors in a particular city.

(ii) Display female dons with particular group.

(iii) Search and display the address of a particular donor, given the name in a text box.

---------- Cut here ----------

4. (a) Write a VB project that stores 10 employee records with fields : empho, name, age, sex, salary in an array. Display data fields in text boxes and provide command buttons to move to desired record.

(b) Write a VB project using built in AX control (Rich Text Box) develop the windows NOTEPAD like editor with file and edit menus and also display the floating menu whenever necessary.

---------- Cut here ----------

5. (a) Write a VB program using control arrays that creates a scientific calculator with appropriate command buttons.

(b) Write a VB program to compute and print either the sum of odd numbers or even numbers at the user's choice using Label, Text and Option buttons.

6. (a) Write a VB project for commercial bank operations using SB account database with the following features :

   (i) Add new account

   (ii) Deposit amount

   (iii) Withdraw amount (minimum balance condition)

   (iv) Close the account.

   (b) Design and implement a digital clock using ActiveX control.

DE–447

Sub. Code
31

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2017.

Third Semester

DISCRETE MATHEMATICS

(2005 to 2010 Calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Show that $\left(\neg P \wedge \left(\neg Q \wedge R\right)\right) \vee \left(Q \wedge R\right) \vee \left(P \vee R\right) \Leftrightarrow R$

   (b) Obtain the principal disjunction normal form of $P \rightarrow \left(\left(P \rightarrow Q\right) \wedge \neg\left(\neg Q \vee \neg P\right)\right)$.

2. (a) Show that $R \wedge \left(P \vee Q\right)$ is a valid conclusion from the premises $P \vee Q, Q \rightarrow R, P \rightarrow M$ and $\neg M$.

   (b) Show that $\left(x\right)\left(P(x) \vee Q(x)\right) \Rightarrow (x)P(x) \vee \left(\exists x\right)Q(x)$.

3. (a) What are the operations performed on set? Explain with examples.

   (b) If $A = \{\alpha, \beta\}$ and $B = \{1,2,3,\}$, what are $A \times B, B \times A, A \times A, B \times B$ and $\left(A \times B\right) \cap \left(B \times A\right)$?

4. (a) What do you mean by an equivalence relation? Explain with example.

(b) Let R and S be two relations on a set of positive integers I.

$R = \{\langle x, 2x \rangle / x \in I\}$

$S = \{\langle x, 7x \rangle / x \in I\}$

Find $R \circ S, R \circ R, R \circ R \circ R$ and $R \circ S \circ R$.

5. (a) What are the types of function? Explain with examples.

(b) Let $f : R \to R$ and $g : R \to R$, where R is the set of real numbers. Find $f \circ g$ and $g \circ f$, where $f(x) = x^2 - 2$ and $g(x) = x + 4$. State whether these functions are injective, surjective and bijective.

6. (a) What are the general properties of algebraic system? Explain.

(b) Show that the set N of natural numbers is a semigroup under the operation $x * y = \max\{x, y\}$. Is it a monoid.

7. (a) State and prove Lagrange's theorem.

(b) Define the following with examples

(i) Undirected graph.

(ii) Subgraph.

(iii) Simple graph.

(iv) Degree of the graph.

(v) Path of the graph.

8. (a) Write the Warshall algorithm to find the path matrix of the graph and give one example. (10)

(b) Define any five terminologies related to trees with examples. (10)

DE–448

Sub. Code
32

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2017.

Third Semester

OPERATING SYSTEMS

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. Narrate the evolution of operating systems in detail. (20)

2. Elucidate the procedure of following operating system structures :

   (a) System calls (12)

   (b) Virtual machines. (8)

3. Write notes on :

   (a) Inter process communications. (10)

   (b) Semaphores concept with the dining-philosophers problem. (10)

4. Discuss the following process scheduling policies with neat diagram :

   (a) Priority scheduling. (8)

   (b) Shortest job scheduling. (6)

   (c) First come first serve scheduling. (6)

5. (a) Write about device controllers. (10)

   (b) Narrate the role of device independent I/O software and its applications. (10)

6. Summarize the concept of deadlock avoidance with banker's algorithm. (20)

7. (a) Discuss the concept of multiprogramming with variable partitioning technique. (10)

   (b) List out various page replacement algorithms and explain any two of it. (10)

8. What are the different kinds of allocation techniques used as disk space management? Explain in detail. (20)

DE–449

Sub. Code
33

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2017.

Third Semester

RDBMS

(2005 to 2010 Calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Define the following :

   (i) data base

   (ii) primary key

   (iii) secondary key

   (iv) foreign key. (10)

   (b) What are the advantages of database? (10)

2. (a) Give the guidelines for preparing a good schema and subschema. (10)

   (b) Discuss the manipulative capabilities available in DBMS. (10)

3. (a) How to navigate the relational model? (10)

   (b) Discuss on data definition facilities available in a relational model. (10)

4. (a) Explain the technique of improving efficiency in a relational model. (10)

   (b) What are the data control facilities available in a relational model? Discuss them. (10)

5. (a) Mention the properties of relational database. (10)

   (b) How networking system is taken care in oracle? (10)

6. (a) Describe the architecture of oracle database. (10)

   (b) Discuss the logical structures in oracle database. (10)

7. (a) How queries are treated in oracle? Discuss on sub queries. (10)

   (b) Explain the technique of selecting distinct values through queries. (10)

8. (a) With examples, show how a table can be modified and renamed. (10)

   (b) Mention the salient features of embedded SQL. (10)

DE–450

Sub. Code
34

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2017

Third Semester

Computer Applications

UNIX AND SHELL PROGRAMMING

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain the basis of files in Unix.

   (b) Describe the various options in is command.

2. (a) What is meant by filter? Explain any two filters.

   (b) What are awk built-in functions? Explain.

3. (a) Describe the features of Meta characters and shell variables.

   (b) Explain the following shell commands with examples.

      (i) pwd

      (ii) who.

4. (a) Discuss the while loop in Unix with example.

   (b) Explain the conditional expression in Unix.

5. (a) What are shell built-in variables? Explain.

   (b) Explain the usage of news and trap commands.

6. (a) Write a Shell program to compare two files.

   (b) What are the various system calls for file creation? Explain.

7. (a) Explain the usage of Iseek and inodes.

   (b) Describe the use of commands get and put in shell programming with example.

8. (a) Explain any ten common ms formatting commands with examples.

   (b) Write a procedure for fibonacci series.

DE–451

Sub. Code
35

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2017.

Third Semester

COMMUNICATION SKILLS

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain the steps in communication process.

   (b) Outline the requisites of effective report writing.

2. (a) Explain the techniques of reading.

   (b) Explain the procedure for Agenda writing.

3. (a) Explain various listening tests.

   (b) What are the features of ideal conversation?

4. (a) What are the methods of asking questions?

   (b) Explain the various types of interview.

5. (a) Explain the process of group discussion.

   (b) How is body language used as a tool for personality identification? Explain.

6. (a) Explain the techniques for negotiation.

   (b) What are the desirable features of interpersonal behaviour?

7. (a) Explain the features of an ideal meeting.

   (b) Explain the importance of communication in management.

8. (a) Explain the Union's role in communication.

   (b) Explain the requisites of an ideal presentation.

DE–9536

Sub. Code
36

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2017.

Third Semester

Lab V — RDBMS

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

One Question Should be given to each student by LOT system.

1. (a) An examination has been conducted to a class of 10 students and the following are given as inputs :

Three test marks, regno, name.

Write a program to do the following :

Assign a letter grade to each student based on the average score and list the register number, average score and grade. (Assume your own grade system)

(b) Write a PL/SQL program to evaluate DA, HRA, Income Tax, Grosspay and Netpay. Base Table PAY contains : Empno, name.

basic pay, DA, HRA, IT, GPAY, NPAY

Input from the base table is empho, name, basic pay.

2. (a) Write a program for hospital billing system with the following fields :

Patient no., name, age, attending doctor, patient type (In/out), consulting charge, blood test charge and total fee.

Prepare reports with the following conditions :

(i) The patients undergone blood test.

(ii) The patients who belongs to in patient category.

(b) Write a PL/SQL program to search the guide name for the given student regno.

Student tables contains : regno, title, name, guide name.

---------- Cut here ----------

3. (a) A company XYZ Ltd., states monthly salary to its employee. It consists of basic salary, allowance, deduction.

DA = 53% of basic salary

HRA = 10% of basic salary

Deductions : PF = 10% of basic salary

LIC = payable by an employee

Loan = payable by an employee

Create a master table of 10 records with the fields :

empno, name, designation, basic salary, HRA, DA, bank a/c no, LIC premium no.

A transaction file contains :

Empno, PF subscription, LIC, Loan write a program to prepare a report in neat format.

(b) Write a PL/SQL program to prepare a electricity bill. (Assume your own data).

4. (a) A daily sales file contains records with the following fields :

Deptno, date, item description, sale price for each item, quantity, cost of each item.

Write a program using oracle to list all the input data and compute total amount of sales and profit for each sale.

(b) Write a PL/SQL program to prepare mark statement of a student using student table and mark table. Assume your own data.

---------- Cut here ----------

5. (a) A hospital maintains blood donors record information in a file. The file contains :

donor number, name, age, address, place of birth, blood group (A, AB, B, O)

Write a program to print out the number, name and address of the donors for the following categories.

(i) Blood donors having A blood group.

(ii) Female donors with AB group in the age between 20 and 25.

(b) Create a table containing phone number, name and address. Write a PL/SQL program to search for address using phone number.

DE–9537

Sub. Code
37

DISTANCE EDUCATION

M.C.A./M.C.A (Lateral) DEGREE EXAMINATION,
MAY 2017.

Third Semester

Computer Applications

COMPUTER LAB – VI : ALGORITHM AND SHELL PROGRAMMING

(2005 to 2010 Calendar year)

Time : Three hours Maximum : 100 marks

Examiner has to select and give ONE question to each candidate by LOT system.

1. (a) Write an algorithm and a program for exchange sort.

   (b) Write and execute a shell program to send mail to groups of users by extracting their ids from / etc / group file.

---------- Cut here ----------

2. (a) Write an algorithm and a program for selection sort.

   (b) Write and execute a shell program to print monthly payslips for employees in an organization. Assume your own data.

---------- Cut here ----------

3. (a) Write an algorithm and a program for heap sort.

   (b) Write a program for four function calculator by using yacc command.

4. (a) Write an algorithm and a program for Travelling salesman problem using branch-and-bound method.

(b) Write a shell program to prepare electricity bill for domestic consumers and print the bill in a neat format. (Assume your own data)

---------- Cut here ----------

5. (a) Write an algorithm and a program to find the factorial of a given number using recursion.

(b) Write and execute a shell program to generate leave status report of an organizaiton.

---------- Cut here ----------

6. (a) Write an algorithm and a program for Towers of Honai Problem.

(b) Write a menu driven shell program to copy, edit, rename and delete a file.

---------- Cut here ----------

7. (a) Design and develop an algorithm for knight tour problem. Also write the program.

(b) Write a shell program using 3 arguments to take the pattern as well as input and output file names. If the pattern is found display "pattern found" else display "Error Message". Also check if right number of arguments are entered.

8. (a) Design and develop an algorithm for binary search. Also write the program.

(b) Write a menu driven shell program for the following :

(i) list of files

(ii) processes of users

(iii) today's date

(iv) users of system

(v) quit to unix.

DE–452

Sub. Code
41

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2017.

Fourth Semester

COMPUTER NETWORKS

(2005 to 2010 Calender year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Explain transmission media with necessary diagrams. (10)

   (b) Compare and contrast ISDN and ATM. (10)

2. (a) Discuss the error detection and correcting codes. (10)

   (b) Describe the sliding window protocols. (10)

3. (a) Write down the network layer design issues and its functions. (12)

   (b) Write a note on internet multicasting. (8)

4. (a) Explain in detail about the internet transport protocols. (10)

   (b) Discuss about flow control and buffering. (10)

5. (a) Give a detailed note on cryptography. (10)

   (b) Discuss in detail about world wide web. (10)

6. (a) Write short notes on telephone system. (10)

   (b) Explain the design issues of data link layer. (10)

7. (a) Give a brief note on firewalls. (10)

   (b) Discuss shortly about multiplexing. (10)

8. (a) Describe any two routing algorithms. (10)

   (b) Explain about the measuring of network performance. (10)

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Sub. Code
42

DISTANCE EDUCATION

M.C.A/M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2017.

Fourth Semester

ACCOUNTING AND FINANCIAL MANAGEMENT

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

(5 × 20 = 100)

1. Rectify the following errors by using Suspense Account :

   (a) Sales day book was overcast by Rs. 100

   (b) A sale of Rs. 50 to *x* was wrongly debited to *y* account

   (c) General expenses Rs. 18 was posted in the account as Rs. 80

   (d) A bills receivable for Rs. 155 was written in bills payable book. The bill was given by *p*.

   (e) Legal expenses Rs. 119 paid to Raja was debited to his personal account

   (f) Cash received from C. Dass was debited to G. Dass Rs. 150

   (g) While carrying forward the total of one page of purchase book to the next, the amount of Rs. 1,235 was written as Rs. 1,325

   Pass necessary entries for the rectification of about errors and show suspense account.

2. "Analysis without interpretation is meaning less and interpretation without analysis is impossible" – Discuss.

3. The following were the Balance Sheet of Anil Corporation as on 31st December 2000 and 1999. Prepare a source and application of funds statement with a supporting schedule of working capital.

Balance Sheet of Anil Corporation

Liabilities	2000 Rs.	1999 Rs.	Assets	2000 Rs.	1999 Rs.
Accounts payable	12,000	15,000	Cash	45,000	30,000
Bonds payable	60,000	20,000	Account receivable	42,000	50,000
Reserve for depreciation on plant	19,000	10,000	Merchandise inventory	33,000	20,000
Common stock	1,20,000	1,00,000	Plant	1,50,000	1,00,000
Surplus	59,000	55,000			
	2,70,000	2,00,000		2,70,000	2,00,000

The Surplus account on 31st December, 2000 was as follows :

Surplus Account

2000		Rs.	2000		Rs.
June 20	To Cash dividend	12,000	Jan. 1	By Balance b/d	55,000
Dec. 31	To Balance c/d	59,000	Dec. 31	By Net profit for the year	16,000
		71,000			71,000

4. Given the following information for *ABC* company at the end of 1999 determine balances for the income statement and the balance sheet :

Net sales : Rs. 1,00,000

Debtor's turnover ratio based on net sales : 2

Inventory turnover ratio : 1.25

Fixed assets turnover ratio : 0.8

Debt – Assets ratio : 0.6

Net profit margin : 5%

Gross profit margin : 25%

Return on investment : 2%

ABC Company

Income statement (for the year ending Dec. 31, 1999)

Sales	Rs. 1,00,000
Cost of good's sold	----------
Gross profit	----------
Other expenses	----------
Earning before tax	----------
Taxes @ 50%	----------
Earnings after tax	----------

Balance Sheet (as on 31st Dec. 1999)

Liabilities	Rs.	Assets	Rs.
Equity	--------	Net fixed assets	--------
Long term debt	--------	Inventory	--------
Sport term debt	50,000	Debtors	--------
		Cash	--------
Total		Total	

5. From the following information about materials, prepare a material ledger account using last in first out material

2003		
July 1	Opening balance	800 units @ Rs. 2.50
July 8	Purchase of materials	900 units @ Rs. 2.60
July 12	Issue of materials	700 units
July 15	Purchase of materials	600 units @ Rs. 2.70
July 20	Issue of materials	650 units
July 25	Purchase of materials	800 units @ Rs. 2.85
July 27	Purchase of materials	500 units @ Rs. 2.90
July 29	Issue of materials	600 units
July 31	Issue of materials	400 units

6. A factory has three production departments and two service departments. The overhead departmental distribution summary shows the following :

Departments	Rs.
A	6,50,000
B	6,00,000
C	5,00,000
P	1,20,000
Q	1,00,000

The service department expenses are allotted on a percentage basis as follows :

	Production departments			Service departments	
Service department P	30	40	15	–	15
Service department Q	40	30	25	5	–

Show how the expenses of the two service departments are to be charged to production department under "repeated distribution" method and under "simultaneous equation method".

7. A practicing charted accountant now spends Re. 0.90 per kilometre on taxi fares for his client's work. He is considering two other alternatives, the purchase of a new small car or an old bigger car.

The estimated cost figures are :

Items	New Small Car Rs.	Old Bigger Car Rs.
Purchase price	35,000	20,000
Sale price, after 5 years	19,000	12,000
Repairs and servicing (per annum)	1,000	1,200
Taxes and insurance per litre	1,700	700
Petrol consumption per litre	10 km	7 km

Petrol price Rs. 3.50 per litre

He estimates that he does 10,000 km, annually. Which of the three alternatives will be cheaper? If his practice expands and he has to do 19,000 km per annum, what should be the decision? At how many km per annum will the cost of the two cases break-even and why? Ignore interest and income tax.

8. The following information relates to a flexible budget at 60% capacity. Find out the overhead casts at 50% and 70% capacity and also determine the overhead rates :

	Expenses at 50% capacity Rs.
Variable overheads :	
Indirect labour	10,500
Indirect materials	8,400
Semi-variable overheads :	
Repairs and maintenance (70% fixed, 30 variable)	7,000

	Expenses at 50% capacity Rs.
Electricity (50% fixed, 50% variable)	25,200
Fixed overheads :	
Office expenses including salaries	70,000
Insurance	4,000
Depreciation	20,000
Estimated direct labour hours	1,20,000

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Sub. Code
43

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2017.

Fourth Semester

RESOURCE MANAGEMENT TECHNIQUES

(2005 to 2010 Calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

(5 × 20 = 100)

1. Explain the scope and objectives of resource management technique.

2. What are the essential characteristics of linear programming?

3. Old hens can be bought for Rs. 2 each but young ones 5 eggs per week each egg being worth 30 paise. A hen cost's Rs. 1 per week to feed. If a person has only Rs. 80 to spend on the hens, how many each kind should he buy to get a profit of more than Rs. 6 per week assuming that he cannot house more than 20 hens?

4. Determine the initial basic feasible solution to the following problem using row minima method

	To				Availability
From	5	2	4	3	22
	4	8	1	6	15
	4	6	7	5	8
Demand	7	12	17	9	

5. Find the optimal assignment for the assignment problem with the following cost matrix.

	I	II	III	IV
A	5	3	1	8
B	7	9	2	6
C	6	4	5	7
D	5	7	7	6

6. A barber with one man takes exactly 25 minutes to complete over hair cut. If customer arrives in a poisson fashion at an average rate of one every 40 minutes. How long on the average must a customer wait for service?

7. The cost of new machine is Rs. 4,000. The maintenance cost of nth year is given by $R_n = 500(n-1), n = 1,2....$ suppose that the discount rate per year is 0.05. After how many years it will economical to replace the machine by a new one?

8. A prize distribution function is to be organized the data has been fixed with the chief guest prepare. W.B.S. and draw precedence net work. Also number the events.

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Sub. Code
44

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2017.

Fourth Semester

OBJECT ORIENTED ANALYSIS AND DESIGN

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. Discuss in detail the evolution of the object model. (20)

2. (a) Explain abstraction with an example. (10)

   (b) Explain the different types of inheritance with examples. (10)

3. Write an overview about the following with respect to classes.

   (a) Interface of a class (5)

   (b) Association (5)

   (c) Aggregation (5)

   (d) Instantiation. (5)

4. (a) Explain how a method is invoked. (5)

   (b) Explain the classical and modern approaches of identifying classes and objects. (15)

5. (a) Discuss about the UML meta model. (10)

   (b) Explain the purpose of analysis and design. (10)

6. (a) Write in detail about view process. (10)

   (b) Discuss about use case diagram. (10)

7. (a) Draw the class diagram to explain generalization notation. (10)

   (b) What are the essentials of class diagram? (10)

8. Draw the state diagram and activity diagram for a banking process. (20)

DE–456

Sub. Code
45

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION,
MAY 2017.

Fourth Semester

INTERNET PROGRAMMING

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) What is Internet programming? Discuss the overview of Internet programming. (10)

   (b) Discuss the Design issues of WWW. (10)

2. (a) What is Encryption? Explain how it will be secure in Internet Application. (10)

   (b) Briefly explain the features of Java. (10)

3. (a) Briefly discuss common gateway Interface. Give example CGI Script. (10)

   (b) Write a brief note on VB script. (10)

4. (a) Define HTM. How HTML tags are useful for creating web document? Explain them. (10)

   (b) Write a VB script code to check the username and password. (10)

5. (a) What is browser? Explain Microsoft Internet Explorer. (10)

   (b) Discuss about alignment tag, list tag in HTML. Give example coding. (10)

6. (a) Design a webpage using Active X controls. (10)

   (b) Explain VDolive technology. (10)

7. (a) How do you create a custom Integrated Application using Multiple protocols? (10)

   (b) What is hyperlinks? Explain briefly. (10)

8. (a) How do you implement graphics in a webpage? Demonstrate with an example coding. (10)

   (b) Write a brief note on Animation with suitable example. (10)

DE–9538

Sub. Code
46

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2017.

Fourth Semester

LAB VII — INTERNET PROGRAMMING

(2005 to 2010 Calendar year)

Time : Three hours Maximum : 100 marks

Examiner has to select and give ONE question to each candidate by lot system.

1. (a) Write a Java program to create a color as the background.

   (b) Write a Java program to display the dialogue menu using applet.

---------- Cut here ----------

2. (a) Write a Java Script to create a color Palet and display the text in the color chosen from the Palet with proper background color.

   (b) Develop a single page advertisement for a shop to be opened shortly.

---------- Cut here ----------

3. (a) Write a VB script to create a calendar for given month and year.

   (b) Write a swing program to create the tabbed panels.

4. (a) Write a Java Script to create a window by using the confirm message.

   (b) Develop a picture gallery having atleast five pages. Each of them is having several pictures with suitable information.

---------- Cut here ----------

5. (a) Write a VB Script to do the following.

   (i) Find the difference between the two dates.

   (ii) Find the age of a person by providing date of birth.

   (b) Write a Java program to create the frames and its controls using applet.

---------- Cut here ----------

6. (a) Write a Java Script to create a order form to select the house articles.

   (b) Design and publish a web page for your college.

---------- Cut here ----------

7. (a) Write a VB Script to do the following :

   (i) Check the given password.

   (ii) Change the existing password.

   (b) Write a program to display the different colors and fonts using applet.

---------- Cut here ----------

8. (a) Write a menu driven program to do database functions using database connectivity facility. The functions include display, edit, insert and delete a record from a table.

   (b) Design a single page advertisement for a shop to be opened shortly.

DE–9539

Sub. Code
47

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2017.

Fourth Semester

Lab–VIII – NETWORK LAB

(2005 to 2010 Calender Year)

Time : Three hours Maximum : 100 marks

Examiner has to select and give ONE question to each candidate by LOT system.

1. (a) Write a java program to setup echo server and echo client.

   (b) Write a program to read a file from the remote system.

---------- Cut here ----------

2. (a) Write a Java program to setup a chat server and chart client.

   (b) Write a Java program to send message from one system to another system.

---------- Cut here ----------

3. (a) Write a Java program to find factorial of given number using FACTORIAL INTERFACE technique.

   (b) Write a Java program to setup time server and time client.

4. (a) Write a Java program to find the factorial of a given number using RMI client/server technique.

   (b) Write a Java program to encrypt the given word.

---------- Cut here ----------

5. (a) Write a Java program to multiply two numbers using RMI client/server technique.

   (b) Write a Java program to display the users and their IP addresses logged in the network.

---------- Cut here ----------

6. (a) Write a Java program to add two numbers using RMI client/server technique.

   (b) Write a Java program to setup time server and time client.

DE–457

Sub. Code
51

DISTANCE EDUCATION

M.C.A.(S)/M.C.A.(Lateral) DEGREE EXAMINATION, MAY 2017.

Fifth Semester

IMAGE PROCESSING AND ANALYSIS

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Discuss about the perception of colour. (10)

   (b) Explain basic transformations of geometric model in image formation. (10)

2. (a) Discuss about the properties of perspective projection in geometric model. (10)

   (b) Explain Fuzzy mathematical recognition. (10)

3. (a) Discuss about edge preserving smoothing. (10)

   (b) Write a detailed note on restoration by Singular value decomposition. (10)

4. (a) Explain the restoration by homomorphic filtering. (10)

   (b) Discuss about any one Lossy compression technique. (10)

5. (a) Explain stereo imaging Algorithms. (10)

   (b) Discuss about the Threshold selection methods. (10)

6. (a) How to determine an optimum threshold for gray level thresholding? (10)

   (b) Write a note on pattern fitting approach. (10)

7. (a) Discuss about geometric attributes. (10)

   (b) Explain in detail about statistical classification. (10)

8. (a) Write a detailed note on region based description. (10)

   (b) Describe the elements of digital geometry. (10)

DE–458

Sub. Code
52

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2017.

Fifth Semester

SOFTWARE ENGINEERING

(2005 to 2010 Calendar year onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain the software applications. (10)

   (b) Discuss about the different versions of spiral model. (10)

2. (a) Describe the different types of decomposition techniques. (10)

   (b) How to plan programming team structure? Explain. (10)

3. (a) Explain the software configuration management process. (10)

   (b) Discuss about the two distinct types of risks. (10)

4. (a) What do you mean by software requirements specification? Explain. (10)

   (b) Write a note on information flow. (10)

5. (a) What is integration testing? Explain. (10)

   (b) Write a note on cohesion and coupling. (10)

6. (a) List and explain the types of software resources. (10)

   (b) Explain the factors that needed for software project planning. (10)

7. (a) Write a note on quality standards. (10)

   (b) Explain the components of software requirements gathering. (10)

8. (a) Discuss about software design concepts. (10)

   (b) Describe the interface design guidelines. (10)

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Sub. Code
53

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2017

Fifth Semester

VISUAL PROGRAMMING — II

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) What is Hungarian Notation? Explain. (10)

   (b) Explain the following :

      (i) Main frame object. (5)

      (ii) Document object. (5)

2. (a) What are mouse events? Explain with an example. (10)

   (b) How to add a menu item in code? Give in example coding. (10)

3. (a) Write a program to create a simple calculator. (10)

   (b) Where do you use list and combo boxes? Explain through programs. (10)

4. (a) How can you read and write files? Explain. (10)

   (b) What is Serialization? Explain with examples. (10)

5. (a) Explain the following :

   (i) Scroll Bars. (5)

   (ii) Splitter windows. (5)

   (b) What is exception handling? How can you catch multiple exceptions? (10)

6. (a) Write a VC++ program to create employee database. (10)

   (b) Explain the concept of building Active X container application. (10)

7. (a) What is MDI? Explain with an example. (10)

   (b) Write a VC++ program to find whether the given number is prime or not. (10)

8. (a) Explain sequential and random access files. (10)

   (b) How can you sort and filter records in a database? Explain. (10)

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Sub. Code
54

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2017.

Fifth Semester

COMPILER DESIGN

(2005 to 2010 CY onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain the structure of a compiler. (10)

   (b) Write a note on regular expressions. (10)

2. (a) Describe optimization. (15)

   (b) Discuss about error handling. (5)

3. (a) Give a detailed note on shift reduce parsing. (10)

   (b) How to construct LALR set of items. (10)

4. (a) Explain top-down parsing. (10)

   (b) Discuss about LR parsers. (10)

5. (a) Explain the implementation of syntax directed translation. (10)

   (b) Discuss about postfix notation. (10)

6. (a) Describe three address code. (10)

   (b) Discuss about post fix translations. (10)

7. (a) Explain the contents of a symbol table. (12)

   (b) Give a note on errors. (8)

8. (a) Explain loop optimizations. (10)

   (b) Describe DAG representation of basic blocks. (10)

DE–461

Sub. Code
55

DISTANCE EDUCATION

M.C.A (S)/M.C.A (S) (Lateral) DEGREE EXAMINATION, MAY 2017.

Fifth Semester

DATA WAREHOUSING AND MINING

(2005 to 2010 cy)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Give a more detailed account of the stages of the KDD process. (10)

   (b) Write a note on client/server and data warehousing. (10)

2. (a) What is k-nearest neighbour? Explain. (10)

   (b) Discuss about the different forms of neural network. (10)

3. (a) List and explain the steps for designing summary tables. (10)

   (b) Explain the architecture of a ware house manager. (10)

4. (a) List and explain any three server hardware architecture. (10)

   (b) Discuss about the actual backup and restore strategy. (10)

5. (a) What is tuning query? Explain. (10)

   (b) Write a note on database sizing. (10)

6. (a) What is dimension data? Explain. (10)

   (b) Explain the steps and guidelines for determine facts from dimensions. (10)

7. (a) Discuss about the database layout. (10)

   (b) Explain about data warehouse features. (10)

8. (a) Write a note on adhoc queries. (10)

   (b) Explain the tasks in developing test plan. (10)

DE–9540

Sub. Code
56

DISTANCE EDUCATION

M.C.A/M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2017

Fifth Semester

LAB IX — VISUAL C++

(2005 to 2010 Calendar year)

Time : Three hours Maximum : 100 marks

Examiner should select and give ONE question to each candidate by lot system.

Break Up of Marks

Record Note Book	:	5
Algorithm/Flowchart	:	15
Program	:	35
Debugging and Execution	:	35
Result	:	10
Total marks	:	100

1. (a) Write a visual C++ win 32 application program using MFC that displays a message "Hello".

   (b) Write a visual C++ win 32 application program using MFC that creates two push buttons OK and CANCEL on the client area. Buttons should respond to user click over them and display appropriate message.

2. (a) Write a visual C++ win 32 application program using MFC that creates a new font.

   (b) Write a visual C++ win 32 application program using MFC that displays the status of ALT, CTRL, SHIFT, NUM LOCK and SCROLL LOCK keys.

---------- Cut here ----------

3. (a) Write a visual C++ win 32 application program using MFC that creates a list box and displays name of the states in India

   (b) Write a visual C++ win 32 application program using MFC that displays current mouse coordinates in status bar.

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4. (a) Write a visual C++ win 32 application program using MFC that fills the back ground of the client area with a bitmap.

   (b) Write a visual C++ win 32 application program using MFC that displays "Good Morning" wherever the user clicks the mouse button on the client area.

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DISTANCE EDUCATION

M.C.A./M.C.A (Lateral) DEGREE EXAMINATION, MAY 2017.

Fifth Semester

LAB X — Compiler Design

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Examiner should select and give ONE question to each candidate by LOT system.

1. (a) Write a program in C to recognize identifier in the given source code.

   (b) Write a C program to construct SLR parsing table for the following grammar.

   $E \rightarrow E + T/T$

   $T \rightarrow T * F/F$

   $F \rightarrow (E)/id$

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2. (a) Write a C program to recognize constant in the given sources code.

   (b) Write a C program to construct of a canonical LR parsing table for the following grammar.

   $S \rightarrow CC$

   $C \rightarrow cC / d.$

3. (a) Write a C program to convert infix expression to postfix notation.

(b) Write a C program to construct predictive parsing table for the following production rule:

$S \rightarrow$ iCt SSI/a

$SI \rightarrow$ es/ E (E is empty)

$C \rightarrow$ b

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4. (a) Write a C program to draw the transition diagram for DFA by accepting the following regular expression:

(i) $(a/b)^* \; abb$

(ii) $(1/0)\,1+.10\;(1/0)\,*$.

(b) Write a C program to translate the expression $(a+b)*(b-c)/(a+b)$ into post fix notation and three address code, quadraples, triples and Indirect triples.

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5. (a) Write a program to remove spaces, comment lines, new lines and produce line numbers. (preliminary scanning).

(b) Construct a C program to accept a string and check whether it satisfies the following production rule (By Using Operator precedence grammar).

$S \rightarrow AbB$

$A \rightarrow aBb$

$B \rightarrow a/Bba.$

6. (a) Write a C program to accept a string and check whether it satisfies the following production rule (By using shift reduce parsing).

$E \rightarrow E + E \,/\, E - E \,/\, E * E \,/\, E \,/\, E \,/\, id \,/$ Const/(E)

(b) Write a C program to construct predictive parsing table for the following production rule:

$S \rightarrow$ i C t SSI/a

$SI \rightarrow$ eS/E (E is empty)

$C \rightarrow b$.

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DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2017.

Sixth Semester

SOFTWARE PROJECT MANAGEMENT

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Define software projects and compare other types of projects. (10)

   (b) Give the overview of project plan with neat flowchart. (10)

2. (a) Write notes on project evaluation. (10)

   (b) Explain : Software estimation. (10)

3. (a) What are the activity plannings? Explain. (10)

   (b) Discuss network planning model. (10)

4. (a) What is risk? Discuss risk evaluation. (10)

   (b) Explain resource allocation and their characteristics. (10)

5. (a) Illustrate monitoring and control management. (10)

   (b) Discuss leadership. (10)

6. (a) How to organize people and teams? Explain. (10)

   (b) How do plan for small projects? (10)

7. Explain in detail about SCM. (20)

8. (a) Illustrate the standards of configuration management. (10)

   (b) Write about prototyping. (10)

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DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2017.

Sixth Semester

MULTIMEDIA SYSTEMS

(2005 to 2010 Calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Describe the benefits of using multimedia. (10)

   (b) Explain the types of development tool. (10)

2. (a) Explain the functions and subsystems of system components. (10)

   (b) Discuss about commercial tools. (10)

3. (a) Explain Text representation and Text operation. (10)

   (b) Discuss about animation operation. (10)

4. (a) Explain Image Operations. (10)

   (b) Discuss about representation of music. (10)

5. (a) Write about CD-ROM. (8)

   (b) Describe CD-i Architecture. (12)

6. (a) Explain CD-i Media types. (10)

   (b) Discuss CD-i operations. (10)

7. (a) Describe objects and classes. (8)

   (b) Explain component classes. (12)

8. (a) Discuss about Image compression techniques. (12)

   (b) Explain MPEG standards. (8)

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DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2017.

Sixth Semester

NEURAL NETWORKS

(2005 to 2010 CY)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) What is biological neural network? Explain. (10)

   (b) Explain the applications of artificial neural networks. (10)

2. (a) Define Learning. What are the different types of learning rules? (6)

   (b) Discuss in detail back propagation networks. (14)

3. (a) Differentiate single and multilayer concepts. (8)

   (b) Explain counter propagation networks. (12)

4. (a) Illustrate Hopfield network with necessary diagrams. (12)

   (b) Discuss associative memory. (8)

5. (a) What is Bidirectional Associative Memory (BAM)? Explain. (10)

   (b) Explain adaptive and competitive with necessary examples. (10)

6. (a) Discuss adaptive resonance theory. (8)

   (b) Discuss in detail optical neural network. (12)

7. (a) Explain about holographic correlators. (12)

   (b) Differentiate cognition and neo-cognition. (8)

8. (a) Draw the architecture of ART network and explain in detail. (12)

   (b) State the need for training the Neural network. (8)

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DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2017.

Sixth Semester

C SHARP (C#)

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Describe the evolution of C#. (10)

   (b) Discuss .NET frame work with all components. (10)

2. (a) What is meant by VOS? Explain the types of VOS. (10)

   (b) Why do you need CLS? Discuss about CLS. (10)

3. (a) How do you implement the structure concept in C#? Explain with an example. (10)

   (b) What do you understand about Boxing and Unboxing? Explain with an example. (10)

4. (a) Discuss the concept of constructure with an example. (10)

   (b) Why do you need of Indexers? Explain it. (10)

5. (a) Describe the all if control statements in C#. (10)

   (b) When will we apply exception handling? Explain the necessary of exception handling. (10)

6. (a) Explain how to create simple client program. (10)

   (b) Write a note on name space. (10)

7. (a) How do you define a method in C# class? Explain with an example. (10)

   (b) Discuss the concept of events in C#. (10)

8. (a) Explain the inter operating with unmanaged code. (10)

   (b) Discuss about security concept in C#. (10)

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DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2017.

Sixth Semester

LAB—XI : MULTIMEDIA APPLICATIONS

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Examiner should select and give ONE question to each candidate by LOT system.

1. (a) Write a C++ program to gray level transformation using

   (i) Thresholding

   (ii) Bunching.

   (b) Design a birthday greeting page including various graphic features in flash.

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2. (a) Write a C++ program for setting up a pattern for multi-grey-level images or bilevel display using limbs algorithm with the following 2×2 matrix :

$$\begin{matrix} 0 & 2 \\ 3 & 1 \end{matrix}$$

   (b) Do the image animation using flash.

3. (a) Write a C++ program to find the edges of the given image using Sobel edge procedure.

   (b) Prepare a multimedia presentation showing the features of the courses offered in a university.

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4. (a) Write a C++ program to find the best grey level for splitting the given image using automatic selection for splitting technique.

   (b) Design a sales advertisement for a product using flash.

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5. (a) Write a C++ program to compress the given image using contour coding technique.

   (b) Prepare a multimedia presentation shows the features of a software package developed by you.

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6. (a) Write a C++ program for convolution using given template.

   (b) Design an invitation for a function in your college using various graphic features in flash.